LAB EXPERIMENT: LIBRARY MANAGEMENT SYSTEM

1. Aim

To design and develop a simple Library Management System website using HTML and CSS to manage book catalogs, student accounts, book issue and return details, fines, and academic study materials.

2. Algorithm

1. Start the program.
2. Create the basic webpage structure using HTML.
3. Design the page using CSS with suitable colors, layouts, forms, and navigation links.
4. Create a book catalog with book titles, authors, availability status, and search functionality.
5. Create student login and registration forms.
6. Provide a student section to view borrowing history, issue dates, expected return dates, fines, and profile details.
7. Create a librarian dashboard to add, update, and delete books, issue and return books, and calculate fines.
8. Create a study materials section organized by academic year and semester.
9. Add a contact form for student queries.
10. Display the completed webpage in a browser.
11. Stop the program.

3. Code Snippet

HTML and CSS code (index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>Library Management System</title>
  <style>
    body {
      font-family: Arial;
      background: #f2f5f9;
      text-align: center;
    }

    h1 {
      background: #173b57;
      color: white;
```

```
            padding: 20px;
        }

        .box {
            background: white;
            margin: 20px auto;
            padding: 20px;
            width: 300px;
            border-radius: 8px;
        }

        input, button {
            padding: 10px;
            margin: 8px;
        }

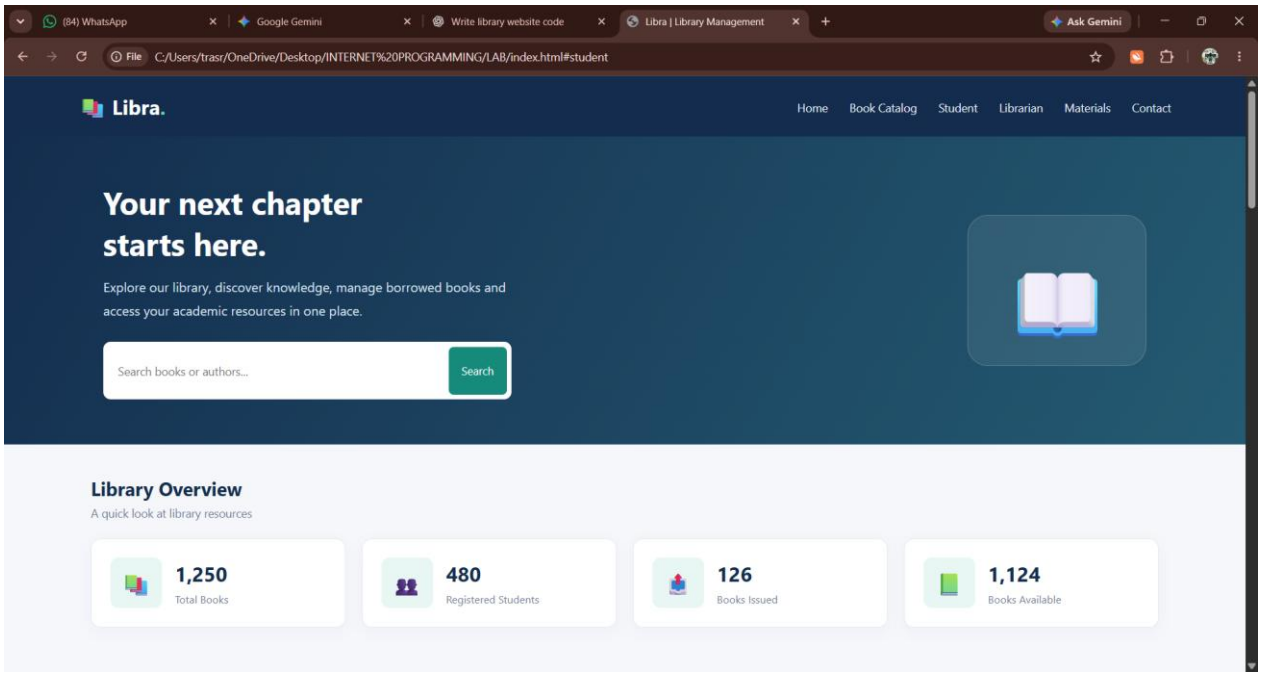
        button {
            background: #168b7c;
            color: white;
            border: none;
            cursor: pointer;
        }
    </style>
</head>
<body>
<h1>Library Management System</h1>
<div class="box">
    <h2>Search Books</h2>
    <input type="text" placeholder="Enter book name">
    <button>Search</button>
</div>

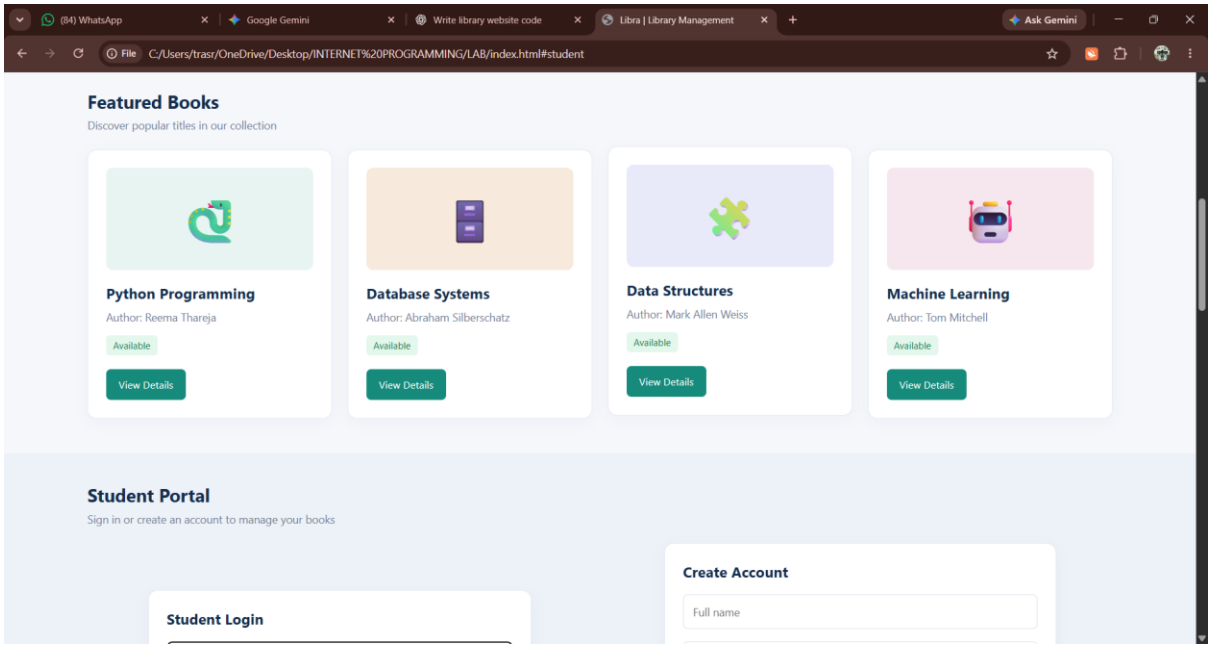
<div class="box">
    <h2>Student Login</h2>
```

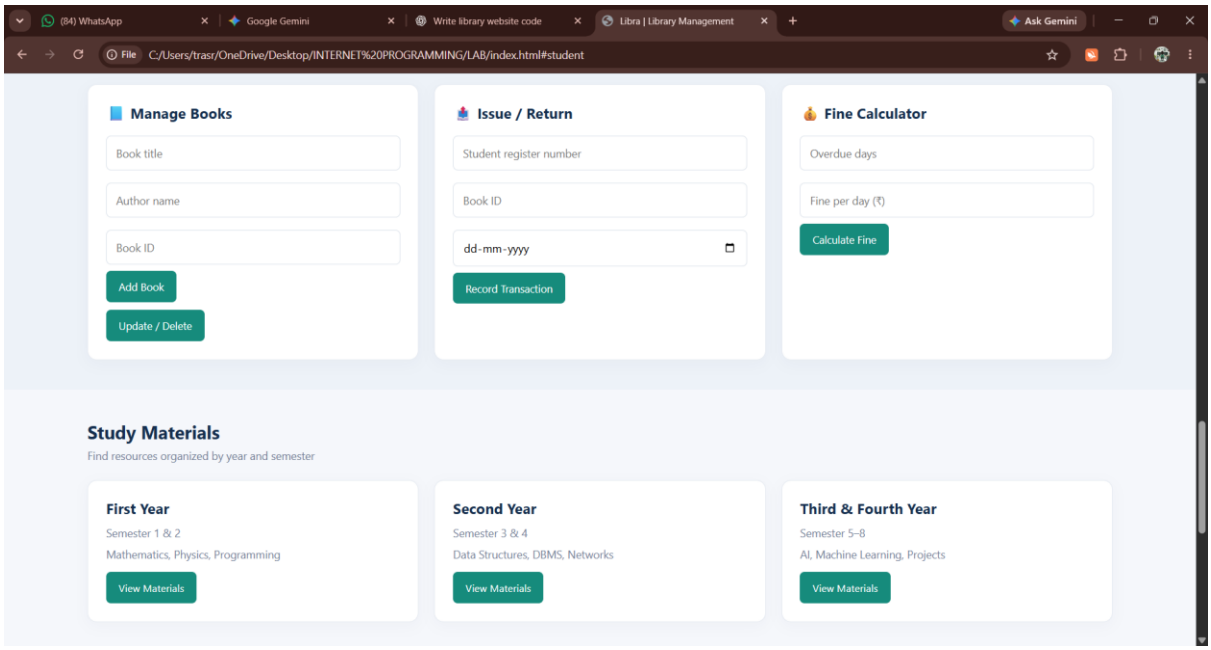
```
    <input type="text" placeholder="Username">
    <input type="password" placeholder="Password">
    <button>Login</button>
</div>
<div class="box">
    <h2>Librarian Admin</h2>
    <button>Add Book</button>
    <button>Issue Book</button>
    <button>Return Book</button>
</div>
<div class="box">
    <h2>Study Materials</h2>
    <p>Year-wise and semester-wise materials</p>
    <button>View Materials</button>
</div>
</body>
</html>
```

4. Output

The webpage displays the following sections:







5. Result

Thus, the Library Management System webpage was designed and developed successfully using HTML, CSS, and basic JavaScript. The interface provides book searching, student forms, librarian management options, fine calculation, and study material sections. Actual account authentication, database operations, and file downloads require backend integration.